

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No: 03

Lab Report on: Linking VS Code with GitHub and Publishing a Live Website.

Course Title : Web Application Lab

Course Code : ICT-3204

Submitted By	Submitted To
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1. Objectives

- To create a GitHub repository and clone it locally in VS Code.
- To understand the Git workflow: add, commit, and push changes.
- To publish a static HTML/CSS/JS website using GitHub Pages.
- To update and maintain a live website through version control.

2. Tools and Technologies

- Visual Studio Code
- Git
- GitHub
- Web Browser
- HTML, CSS

3. Theory

Git is a distributed version control system that allows developers to track and manage changes in their codebase. GitHub provides cloud hosting for Git repositories and enables collaboration.

GitHub Pages is a service that allows developers to host static websites directly from a GitHub repository. It automatically builds and publishes HTML/CSS/JS files to a live URL.

4. Procedure

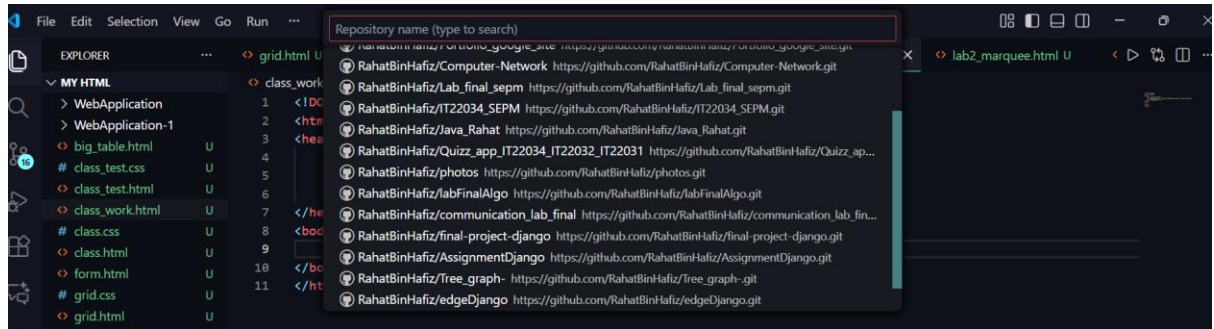
Step 1: Creating a GitHub Repository

1. Log in to GitHub.
2. Create a new repository name Web Application.
3. Keep the repository **public**.

Step 2: Cloning the Repository in VS Code

1. Open VS Code.
2. Open Terminal in VS Code.
3. Run the command:

```
>git clone https://github.com/Sahriar-47/HTML-CSS.git
```



4. Navigate into the cloned repository:

```
>cd Web Application.
```

Step 3: Adding Files

1. Create the website files in the repository folder:

- index.html
- style.css

2. Ensure index.html is **in the root** of the repository.

Step 4: Committing and Pushing Changes

```
>git add .
```

```
>git commit -m "Initial commit with website files"
```

```
>git push
```

Step 5: Publishing with GitHub Pages

1. Go to the repository on GitHub → Settings → Pages.

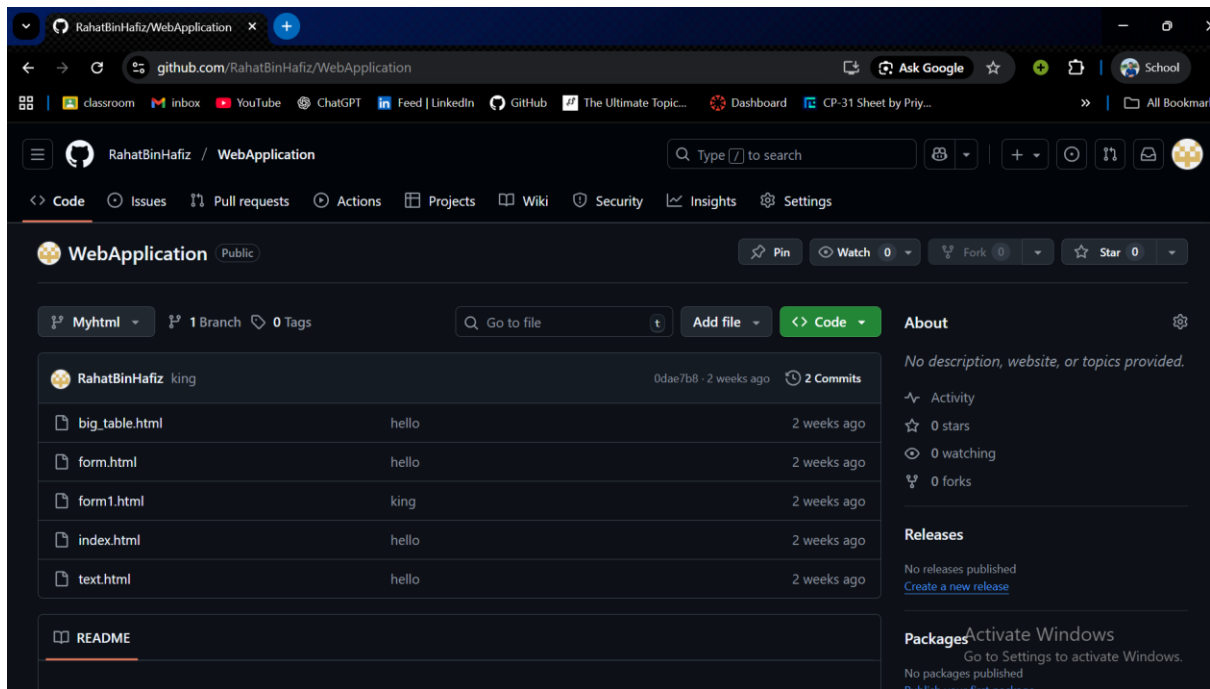
2. Under Source:

- Branch: main
- Folder: /root

3. Click **Save** .

4. Wait 30–60 seconds. GitHub will generate a live URL:

<https://github.com/RahatBinHafiz/WebApplication>



5. Observations

Action

Repository created on GitHub

Cloning repository

Adding website files

Pushing changes

GitHub Pages enabled

Updating files

Result

Successfully created

Local copy available in VS Code

index.html in root, other files added

Changes reflected on GitHub

Live website URL generated

Changes reflected on live site automatically

7. Results

- The static website was successfully published online at:
<https://github.com/RahatBinHafiz/WebApplication>
- Subsequent changes to the website were automatically updated after pushing to GitHub.
- GitHub Pages effectively hosted the HTML/CSS/JS files without requiring a separate server.

8. Discussion

- **Root Folder Requirement:** GitHub Pages requires index.html in the root; otherwise, the live URL will not appear.
- **Repository Visibility:** Pages only works on public repositories unless the user has GitHub Pro.
- **Workflow:** The git add, git commit, git push workflow ensures version control and easy updates.
- **Live Updates:** Any committed changes are automatically reflected on the live website after a short delay.

9. Conclusion

This lab successfully demonstrated how to link VS Code with GitHub and publish a static website using GitHub Pages. The process provides:

- A simple and free method for hosting websites.
- A workflow for maintaining and updating websites efficiently.
- Practical experience with version control, repository management, and live deployment.